

Pratyush Sudhakar

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EDUCATION

Cornell University

Ithaca, NY

Bachelor's of Science in Computer Science (*Minors: Math & Psychology*)

May 2026

Coursework: Adv. ML, Systems Prog., Algorithms (Java), Functional Prog. (OCaml), OOP (Java), Computer Vision

TECHNICAL SKILLS

Languages: Python, Java, TypeScript/JavaScript, C++, SQL, OCaml, HTML/CSS, Bash

Frameworks & Libraries: React, Next.js, Node.js, Express, Flask, Redux, Pandas, Tailwind CSS, Scikit-Learn

Cloud, DevOps & Tools: AWS, GCP, Docker, Kubernetes, Git, PostgreSQL, MongoDB, Redis, Linux, Agile

EXPERIENCE

Computer Systems Developer, Rizvi Lab, Cornell University

July 2024 - September 2024

Tech: Python, Linux/Bash, IoT, Embedded Systems, Git

Ithaca, NY

- Developed PyQt-based touchscreen control interface for BeagleBone systems, enabling real-time monitoring and remote control of food processing equipment through secure SSH connections.
- Automated system operations using Linux shell scripts and cron jobs, reducing manual intervention by 60% and improving equipment uptime monitoring.

Software Engineering Intern, rapStudy

June 2023 - August 2023

Tech: React.js, TypeScript, Redux, Firebase (NoSQL), Material UI, REST APIs

Los Angeles, CA

- Refactored codebase, introduced Redux for state management, and optimized components with memoization.
- Created a secure song-sharing system, using React and 10+ custom Firestore rules for precise access control.
- Developed Redux-powered global audio-playback component, perfectly syncing with lyrics display for enhanced UX.

Data Engineering Intern, Cornell College of Engineering

June 2023 - September 2023

Tech: Python, Pandas/NumPy, Selenium, Next.js, TypeScript, ETL Pipelines

Ithaca, NY

- Engineered a scalable ETL pipeline to process financial data from 3,000+ companies, utilizing Selenium (scraping), NLP, and multithreading, reducing processing time by 83%.
- Built a live data dashboard using Next.js and Server-Sent-Events, cutting latency by 66% with parallelization.

ACTIVITIES

Head of Engineering & President, Perfect Match: Online Matchmaking

January 2023 - May 2025

Tech: Next.js, TypeScript, Tailwind CSS, MongoDB, Redis, AWS, CI/CD

Ithaca, NY

- Directed 4 cross-functional teams through product launch, recruitment, onboarding, and business planning.
- Enhanced UI/UX and set up automated personalized emails (cron jobs), resulting in 600+ new active users.
- Built analytics platform delivering real-time survey insights to 5,000+ users via cached Map-Reduce aggregations, reducing latency by 93%.

Technical Product Manager, Cornell Digital Tech & Innovation

January 2023 - December 2023

Tech: React.js, DynamoDB, Cypress, Kubernetes, Docker, Flutter, Agile

Ithaca, NY

- Spearheaded agile development of cross-platform ride scheduling app with a team of 7 developers.
- Architected test-driven development pipeline using Cypress, Mocha, and Supertest, attaining 94% code coverage.
- Orchestrated CI/CD workflow with GitHub Actions and Kubernetes for streamlined deployment and testing.

VENTURES & PERSONAL PROJECTS

Halo - Wearable AI | Swift (iOS), Python, OpenAI (LLMs), Redis, Celery, IoT

June 2025 - July 2025

Built a real-time face-recognition app (iOS+ESP32) with Apple Vision + OpenAI; refactored code (-40%), trimmed app size (-30%), raised streaming performance (+25%), and cut backend latency from 3 min to 50 s via a Celery/Redis redesign.

AI Tool (Millennium Sponsored) | Flask, Azure, RAG (Vector DB), LangChain

March 2024 - May 2024

Devised a course syllabus assistant with Azure-hosted REST API, utilizing Langchain Models and Zilliz Cluster for vector similarity search, achieving 50ms average response time for context-aware student queries.

Harmonious Sounds | React, Python (Flask), Scikit-Learn, Docker, Nginx

March 2023 - April 2023

Engineered web app with Logistic Regression model predicting Spotify playlist happiness (88% accuracy). Optimized performance using Nginx load balancing across 4 microservices.